

For my project, I'd like to create a 2D model of floating sea-ice in the ocean moving along the surface, and slowly melting due to the exchange of heat between the ocean and the ice. I found a paper from Bernhau et al., 2025, which looks at lab experiments showing how asymmetric ice blocks can propel themselves due to differences in buoyancy from melting. I'd like to simplify this experiment to a perfectly rectangular ice block floating on top of a water surface, and adding my own velocity forcing. The rest of the domain would be a rectangular basin.

My approach would be to couple/implement Navier-Stokes with incompressible and Boussinesq approximation for momentum, and then include temperature and/or salinity transport with the scalar advection-diffusion equation. I would also need to implement ice-melt interactions, which would be through conservation of energy and heat conservation. The sea ice would start off uniformly 0°C, and the water it's on would be maybe 1°C.

With my model, I would run a parameter sweep with ice block width, initial velocity, and ocean temperature and/or temperature difference between the ice and ocean.

References:

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